

A REPORT WRITING
ON
**OPTIMIZING MACHINE WORKSHOP LAYOUT FOR EFFICIENCY AND
PRODUCTIVITY**

IN PARTIAL FULFILMENT OF
MEE 812
DESIGN AND MANUFACTURING OF MACHINE TOOLS

BY
ADELEKE OLORUNNISOLA OYEYEMI: MEE/13/3858

TO
PROF. OLAYINKA AWOPETU

DEPARTMENT OF MECHANICAL ENGINEERING,
SCHOOL OF ENGINEERING,
FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

OCTOBER, 2023

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I. INTRODUCTION

In manufacturing and industrial operations, the significance of machine workshop layout cannot be overstated. Machine workshop layout refers to the systematic arrangement and organization of machinery, equipment, workstations, and resources within a workshop, with the goal of optimizing the workspace for efficiency, productivity, safety, and overall operational performance [1]. This report examines the fundamental principles, quantitative methods, and practical considerations involved in designing an effective machine workshop layout, and applies them to a real case study: the central workshop of the Federal University of Technology, Akure (FUTA).

A. Definition and Significance

Machine workshop layout involves the systematic arrangement of machines, such as lathes, drilling machines, milling machines, power saws, and shaping machines, to facilitate seamless workflow and product manufacturing. The layout is the physical manifestation of a production plan, designed to streamline the movement of materials and personnel while minimizing downtime, errors, and safety risks [2].

A well-structured layout contributes to industrial operations along five dimensions. It directly increases productivity by reducing production-cycle time and maximizing equipment utilization [3]. It improves safety by providing clear pathways, well-marked zones, and hazard-conscious machine placement. It reduces cost by minimizing unnecessary material movement and eliminating layout-induced waste. It provides adaptability, allowing a facility to accommodate new equipment or changing production needs without disruptive reconfiguration. Finally, it improves the working environment: ergonomically designed workstations reduce physical strain and contribute to workforce well-being [4]. Fig. 1 summarizes these five dimensions.

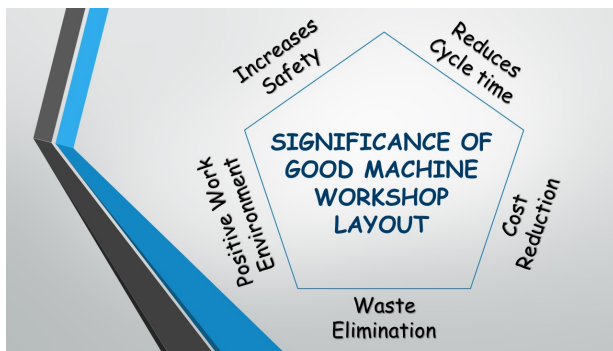


Fig. 1. The significance of a well-designed machine workshop layout spans productivity, safety, cost, adaptability, and workforce well-being.

B. Purpose and Scope

This report has four objectives: to educate readers on the principles of workshop layout design; to inform lay-

out decision-making through theoretical grounding and a worked case study; to demonstrate how layout choices affect efficiency and productivity; and to surface the practical challenges involved in going from a qualitative layout preference to an implementable floor plan.

II. THEORETICAL FRAMEWORK

A. Principles of Machine Workshop Layout Design

Three principles underpin an effective layout. *Flow and workflow*: the logical sequence of operations should be mapped so that materials and workpieces move through the workshop with minimal backtracking, crossing, or unnecessary transport [5]. *Safety and ergonomics*: clear pathways, well-marked zones, and ergonomically designed workstations are not an add-on to the layout but a first-order design constraint [6]. *Space utilization*: the arrangement of machinery should maximize the density of productive workstations per unit floor area without compromising safety or workflow, since floor space is one of the most expensive resources in any facility [1].

B. Lean Manufacturing and the 5S System

Two concepts from Japanese manufacturing practice have particularly strong influence on layout design.

Lean manufacturing is a systematic approach to minimizing waste while optimizing productivity, built around Just-in-Time production, continuous improvement (*Kaizen*), and inventory reduction [7]. Layouts influenced by lean principles favor short distances between successive workstations, minimal material handling, and cellular groupings of machines dedicated to a specific product family, which together reduce the seven classical wastes of manufacturing shown in Fig. 2: overproduction, waiting, transport, overprocessing, inventory, motion, and defects [8].



Fig. 2. The seven wastes (*muda*) of lean manufacturing. A layout that minimizes unnecessary transport and motion directly attacks two of these seven categories.

The 5S system is a workplace organization method comprising five sequential practices, illustrated in Fig. 3: *Sort* (remove unnecessary items), *Set in order* (assign a fixed, logical location to every remaining item), *Shine* (clean and inspect routinely), *Standardize* (codify the first three steps as a repeatable procedure), and *Sustain* (maintain discipline

over the improved state) [9]. A workshop practicing 5S has, by construction, an easier layout-design problem: clutter and undefined storage are two of the most common causes of an ad-hoc, poorly performing layout.



Fig. 3. The 5S cycle: Sort, Set in Order, Shine, Standardize, Sustain.

III. QUANTITATIVE FACILITY LAYOUT DESIGN

The principles above describe *what* a good layout achieves; they do not, by themselves, tell a designer *how* to arrange a specific set of machines. This section introduces the quantitative methodology that the case study in Section VI is evaluated against.

A. Systematic Layout Planning

Muther's Systematic Layout Planning (SLP) procedure is the most widely used structured method for facility layout design [5]. It proceeds from an *Activity Relationship Chart*, in which every pair of departments or work areas is assigned a closeness rating drawn from the coded scale in Table I, to a *relationship diagram* that positions closely-related activities near each other, and finally to a scaled block layout.

TABLE I
MUTHER'S CLOSNESS-RATING CODES USED TO BUILD AN ACTIVITY RELATIONSHIP CHART [5].

Code	Closeness
A	Absolutely necessary
E	Especially important
I	Important
O	Ordinary closeness OK
U	Unimportant
X	Undesirable (should not be adjacent)

B. The From-To Chart and the Layout Objective Function

Where material-flow data is available, layout quality can be stated as a single quantitative objective rather than a qualitative preference. Let f_{ij} be the flow (trips, loads, or mass per period) between department i and department j , recorded in a *from-to chart*, and let d_{ij} be the distance

between their locations in a candidate layout. The standard facility-layout objective is to minimize total material-handling effort,

$$\min Z = \sum_{i=1}^n \sum_{j=1}^n f_{ij} c_{ij} d_{ij}, \quad (1)$$

where c_{ij} is the cost per unit flow per unit distance between i and j , and d_{ij} is typically computed as rectilinear (Manhattan) distance for shop-floor layouts, since material handling equipment generally moves along orthogonal aisles rather than in straight lines [1]. Equation (1) is the formal target that heuristic layout algorithms such as CRAFT (Computerized Relative Allocation of Facilities Technique) and ALDEP (Automated Layout Design Program) attempt to minimize by iteratively exchanging department locations [10].

This is the quantitative benchmark the FUTA case study in Section VI falls short of: closeness ratings there are assigned qualitatively (by educational-workflow reasoning) rather than from a measured from-to chart, a limitation made explicit in Section VI-C.

IV. TYPES OF MACHINE WORKSHOP LAYOUTS

Five layout archetypes recur across manufacturing facilities. Table II summarizes their defining characteristic and best-fit application; the discussion below expands on each, and Figs. 4–7 illustrate the four layouts for which representative diagrams are available.

TABLE II
THE FIVE LAYOUT ARCHETYPES AND THEIR BEST-FIT APPLICATION [1], [3].

Layout	Defining feature	Best-fit application
Fixed	Product stationary, resources move to it	Very large, low-volume products (ships, aircraft, boilers)
Process functional	Machines grouped by operation type	Job shops with high product variety, small-to-medium batches
Line	Machines arranged in a fixed sequence	High-volume, standardized, repetitive production
Cell	Self-contained cells per product family	Frequent product changeover, moderate volume, flexible demand
Combina-tional	Hybrid of the above	Facilities serving both high-volume and customized production

A. Fixed (Position) Layout

In a fixed layout, the product remains stationary while machines, tools, and personnel are brought to it [11]. This is the appropriate layout for very large, low-volume items, such as ships, aircraft, locomotives, and pressure vessels. Its principal advantage is low capital investment, since

machinery is not duplicated or continuously reconfigured; its principal disadvantage is that the work area is often crowded, complicating material storage and handling.

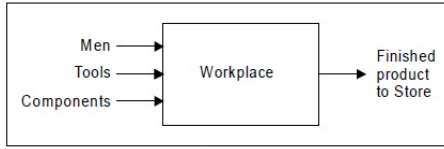


Fig. 2.1 Typical project layout

Fig. 4. Fixed (position) layout: the product remains stationary while resources are brought to it.

B. Process (Functional) Layout

In a process layout, machines are grouped by the operation they perform, for example all shaping machines in one area, all milling machines in another (Fig. 5) [12]. This suits job shops with high product variety, since work-in-progress can be routed through whichever sequence of functional areas a given job requires. The cost of this flexibility is longer, less predictable material-handling routes and a correspondingly more demanding production-control problem.

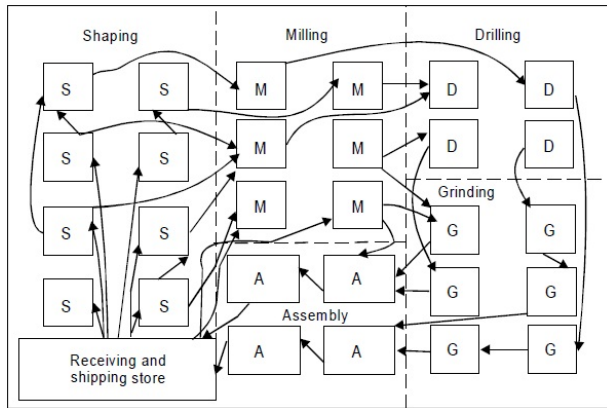


Fig. 2.2 Typical functional layout

Fig. 5. Process (functional) layout: machines are grouped by operation type rather than by product routing [12].

C. Line Layout

A line layout arranges workstations in the sequence dictated by the product's process plan (Fig. 6), so that material moves in one direction with minimal backtracking [13]. This is the layout of choice for high-volume, standardized production, such as automotive assembly. Its efficiency comes at the cost of flexibility: the line's throughput is set by its slowest station, and a design change to the product can require re-engineering the entire line.



Fig. 2.3 Typical line layout

Fig. 6. Line layout: workstations are arranged in the fixed sequence dictated by the process plan [13].

D. Cell Layout

Cell layout groups a subset of machines into a self-contained cell dedicated to a family of similar parts, an application of group technology [3]. Work-in-progress inventory and material travel distance within a cell are both minimized relative to a process layout, and small cross-functional teams often staff a single cell, improving quality feedback. The trade-off is a higher initial investment in re-configurable equipment and cross-trained labor, and cell layouts are generally less space-efficient than a well-balanced line for genuinely high-volume, low-variety production.

E. Combinational (Hybrid) Layout

A combinational layout deliberately mixes fixed, process, and line elements within a single facility, so that a plant can serve both high-volume standardized production and small-batch customized production without maintaining two separate facilities (Fig. 7) [2]. The cost of this versatility is design and coordination complexity: shared resources must be allocated across dissimilar production streams, and workers typically need broader cross-training.

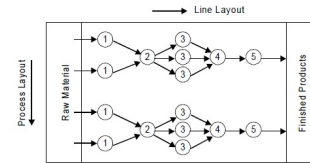


Fig. 2.4 Typical combination layout

Fig. 7. Combinational (hybrid) layout, mixing elements of fixed, process, and line layouts within one facility.

V. SETTING UP A MACHINE WORKSHOP LAYOUT

Fig. 8 summarizes the six-stage procedure followed in this report, adapted from standard facility-planning practice [1].



Fig. 8. The six stages of machine workshop layout design followed in this report.

Initial needs assessment establishes production goals and, where a layout already exists, audits it for bottlenecks and safety concerns. *Workflow analysis* maps the sequence of operations from raw material to finished product and identifies bottlenecks, ideally via a from-to chart as described in Section III. *Space planning* produces the floor plan

itself and assigns zones (production, storage, maintenance, administration) so that dissimilar activities do not interfere with one another. *Ergonomic considerations* shape individual workstation design for comfort, accessibility, lighting, and ventilation. *Safety measures* identify hazards and ensure compliance with occupational safety regulations such as the OSHA general industry standard 29 CFR 1910 [6]. *Technology integration*, finally, anticipates the incorporation of automation, IoT-based condition monitoring, and data analytics into the workshop's operation.

A. Factors Affecting the Layout

Four factor groups most strongly constrain a workshop layout. *Equipment and machinery*: each machine's function, physical footprint, interconnectivity with other machines, and the facility's anticipated future expansion all shape where it can be placed. *Workforce and skill level*: task allocation and ergonomic design must match the skill level and physical needs of the operators who will staff the layout. *Material handling and storage*: flow paths should be short and well-defined, with just-in-time principles reducing the storage footprint required [8]. *Safety regulation and compliance*: occupational safety, machine safeguarding, environmental compliance, and accessibility for inspection are non-negotiable constraints on any candidate layout [6].

VI. CASE STUDY: FUTA CENTRAL WORKSHOP

A. Available Equipment

Table III lists the metal-forming and metal-shaping equipment available at the FUTA central workshop, which forms the basis for the layout proposed in this section.

TABLE III
MACHINES AVAILABLE AT THE FUTA CENTRAL WORKSHOP.

Machine	Number available
Lathe (general)	6
Lathe (CNC)	1
Milling machine (general)	4
Milling machine (CNC)	1
Drilling machine	2
Shaping machine	1
Hydraulic press	1
Horizontal bandsaw	1
Bench work vice	5

B. Proposed Layout and Rationale

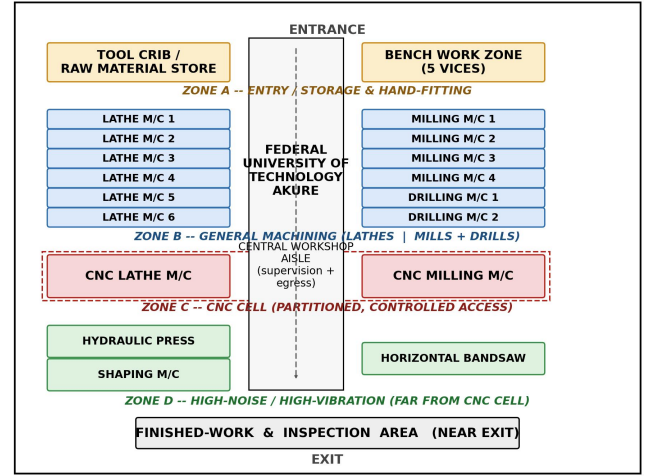


Fig. 9. Proposed zoned layout for the FUTA Central Workshop. Machines are grouped by activity type and vibration/noise sensitivity into four zones along a central supervision aisle running from entrance to exit: Zone A (storage and hand-fitting), Zone B (general machining), Zone C (a partitioned CNC cell), and Zone D (high-noise, high-vibration equipment, positioned as far from the CNC cell as the floor plan allows).

The workshop's purpose is primarily educational rather than high-volume production, which changes the layout objective relative to an industrial job shop: the priority is a layout a student can navigate safely and an instructor can supervise efficiently, not one that minimizes a from-to-chart-derived handling cost. An earlier version of this case study described the proposed layout as a line-process hybrid; that framing does not withstand scrutiny. A line layout implies material moving through a fixed sequence of *different* operations, but this workshop has no fixed product recipe, only multiple redundant machines of the same type (six general lathes, four general mills) used for independent, unrelated student exercises. That is the defining signature of a *process (functional) layout*, not a line, and the report now names it correctly.

Fig. 9 shows the corrected proposal: a zoned functional layout, still built around the central supervision aisle, but with four zones rather than an undifferentiated grouping by machine type alone. *Zone A*, nearest the entrance, holds a tool crib and raw-material store together with a dedicated bench-work area for the five vices, an activity class (hand-fitting, filing, deburring) distinct enough from powered machining that it should not simply be slotted into the milling row, as the original proposal did. *Zone B* groups the six general lathes on one aisle face and the general mills and drills on the other, matching the original's process-layout instinct. *Zone C* is a partitioned CNC cell: the CNC lathe and CNC mill are placed together but separated from the general machining zone, both because CNC equipment is more sensitive to swarf, dust, and vibration from adjacent general machining, and because CNC instruction is typically one-on-one or small-group rather than the simultaneous multi-student use the general-machine rows support. *Zone*

D, at the far end of the aisle from the CNC cell, groups the hydraulic press, shaping machine, and horizontal bandsaw, the workshop's noisiest and highest-vibration equipment, so that their operation does not degrade the precision the CNC cell depends on. A finished-work and inspection area near the exit completes the flow from raw material at the entrance to a finished, inspected part at the exit.

C. Discussion and Limitations

This case study is deliberately presented as a qualitative, adjacency-reasoned layout rather than a from-to-chart-optimized one, and that distinction should be stated plainly rather than implied. The zoning in Fig. 9 reflects closeness ratings in the sense of Table I (an *X*, undesirable-adjacency rating between the CNC cell and Zone D's noisy equipment; an *A* or *E* rating within each general-machining group), but no from-to chart of actual student and material movement between machines was collected to populate Eq. (1), and no CRAFT- or ALDEP-style exchange procedure was run against it. For an educational workshop where batch sizes are small, jobs are heterogeneous, and the walking distance of a student between stations is a pedagogical rather than a cost consideration, this qualitative approach is defensible. It would not be defensible for a production job shop of comparable size, where even a modest from-to chart survey would very likely reveal that grouping by closeness rating alone leaves easily-avoidable material-handling distance on the table. A natural extension of this work is to instrument the FUTA workshop with a short observation period, build an actual from-to chart, and re-evaluate the proposed layout against Eq. (1) rather than against qualitative reasoning alone.

VII. CONCLUSION

Machine workshop layout is not a one-time arrangement decision but an engineering design problem with a well-developed quantitative literature: Systematic Layout Planning, the from-to chart, and the material-handling objective function of Eq. (1) together turn "where should this machine go" into a problem that can be solved rather than merely argued about. The five layout archetypes, fixed, process, line, cell, and combinational, each solve that problem well under a different set of production conditions, and choosing between them is itself an engineering decision driven by production volume, product variety, and the relative cost of space against material handling. The FUTA central workshop case study applies this reasoning to select a zoned functional layout, separating general machining, a partitioned CNC cell, and high-noise equipment, appropriate to an educational setting, while explicitly flagging that the same reasoning, if applied to a production facility, would need to be grounded in measured material flow rather than qualitative closeness ratings alone.

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